⚡ PAIR-PROGRAMMING PROTOCOL • AI CODING STRATEGY

The Agentic Truth Loop: Search to Locate, Read to Know, Quote to Prove, Run to Verify

Why naive AI guesses, and how an engineering agent anchors every claim to raw source disk and live shell execution.

Project: For Each and Every | Region: us-east-2 | Loop: Live & Closed

🔎 1. FIND — STAGE I

Search broadly via grep, symbol index, or vector retrieval. **Take results as candidate leads, never as final answers.**

```
Tool: search_web / run_command(grep)
Output: File paths & line candidates
```

📖 2. READ — STAGE II

Open the raw file on disk. Read surrounding context, exact scalar numbers ($5.00), discrete counts (48), and hard negations ("NOT").

```
Tool: read_file(Path, StartLine, EndLine)
Rule: Never trust cached snippets
```

❝ 3. QUOTE — STAGE III

Answer directly from verified text. Provide exact verbatim quotes, commit SHAs (#a41f89c), and clickable links with line anchors.

```
Format: [File.ext#L10-L20](file:///...)
Standard: Auditable by any reviewer
```

⚡ 4. RUN & VERIFY — STAGE IV

Prove truth through empirical execution: run test suites, check exit codes (0 vs 1), hash payloads (SHA-256), and probe live endpoints.

```
Tool: run_command(pytest/curl)
Proof: Exit 0 • 100% Hash Match
```

🛠️ Which Tool For Which Representation Job? — PRECISION MATRIX

SITUATION / TARGET	ENGINEERING TOOL	WHY (INVARIANT STRENGTH)
Raw Code, Contracts & Configs	`read_file (Raw Disk)`	Zero lossiness. Exact quotes, syntax trees, line numbers, and units preserved.
Rules, Limits, Numbers, Permissions	`Fixed JSON / AST Check`	Rigid schema contracts. Negations ("NOT in us-east-1") and $5 ceilings cannot blur.
Repo Discovery & Conceptual Lookups	`Hybrid Vector Retrieval`	Finds related modules via dense cosine, but rescues negation loss with metadata SQL.
Multi-Hop Connections & Topologies	`Knowledge Graph (Cypher)`	Follows deterministic entity links (12 edges). Every traversed hop points to source truth.
Assertion & Dynamic State Verification	`run_command (Execution)`	Empirical truth: live HTTP 401 gate, S3 74B checksum match, Actions Run ID sync.

⚠️ The 5 Traps & How We Defeat Them — FAILURE TAXONOMY

1. **The Blur Trap (Negations & Numbers)**
   Vectors map "NOT in us-east-1" to us-east-1 at 91.4% similarity. *Defeat:* Enforce hybrid SQL metadata filter.
2. **The Stale Index Trap**
   Code pushed to branch, but test checks old Run #39419075144. *Defeat:* Sync live commit SHA #a41f89c dynamically.
3. **The Ghost Edge Trap**
   Graph claims 20 facts across only 12 edges. *Defeat:* Separate into 12 edge transitions + 8 property contracts.
4. **The Severed Chunk Trap**
   Window split cuts off pronoun from antecedent. *Defeat:* AST-aware structural hierarchy chunking.

🎯 The 5 Truth Checks (After Every Code Change) — CONTINUOUS VERIFICATION

1. **Known-Answer Probes**
   Ask 10–20 invariant questions. Confirm correct passage appears in Top-5.
2. **Bit-for-Bit Hash**
   Compute SHA-256 on raw content. Verify 74B upload matches download 100%.
3. **Poison Negations**
   Inject "NOT" and off-by-one numbers ($5.01). Verify system flags mismatch.
4. **Graph Edge Tracing**
   Sample traversed edges. Assert 100% trace to exact source sentences.
5. **Auth Pre-Flight Knocks**
   Knock 1 (no key ➔ 401), Knock 2 (wrong key ➔ 401), Knock 3 (owner ➔ 200).
6. **Three Monsters Metric**
   Track Missing %, Extra %, Unfindable % separately, never averaged into 1 number.

⚖️ The Golden Law of Agentic Truth — CORE AXIOM

> "A search index is a compass, not the destination. A search hit tells you where to look; only reading raw disk and executing tests proves what is true."

- **Naive Chatbot:** Guesses from frozen weights, trusts search snippets, invents facts.
- **Agentic Assistant:** Locates raw source, reads lines on disk, quotes evidence, and executes verification commands.
- **Human in the Loop:** Verifies high-impact decisions; AI automates the audit trail.